Daily Ritual — Complete Project Report

Project Overview

Name: Daily Ritual
Type: Progressive Web App (PWA)
Purpose: A personal 20-minute daily workout tracker with cloud sync, streak tracking, exercise visualizations, timers, nutrition guidance, and Spotify music integration (partially working)
Live URL: https://nidhithakur-24.github.io/daily-ritual
GitHub Repo: https://github.com/NidhiThakur-24/daily-ritual
Built by: Nidhi Thakur (3rd year AIML student)
Status: Live and functional. Single user. No authentication.

Tech Stack

Layer	Technology	Details
Frontend	Pure HTML, CSS, JavaScript	Single file — index.html. No frameworks, no build tools.
Database	Supabase (PostgreSQL)	Free tier. REST API used directly via fetch().
Hosting	GitHub Pages	Static hosting. Free. Auto-deploys on push to main branch.
Fonts	Google Fonts	DM Serif Display + DM Sans
PWA	Web App Manifest + meta tags	Add to home screen support on Android and iOS
Music	Spotify Web API	Implicit Grant flow. Partially working — login blocked on Edge browser.

Supabase Configuration

Project URL: https://zkavknunbdkmdketrlvg.supabase.co
Anon Key:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6InprYXZrbnVuYmRrbWRrZXRybHZnIiwicm9sZSI6ImFub24iLCJpYXQiOjE3NzM4Mzk2NTYsImV4cCI6MjA4OTQxNTY1Nn0.yf7Evnj29M0TNfuyNhO_mJdAos7ChV5FZZAHrfW39D8

Project ID: zkavknunbdkmdketrlvg
Region: South Asia (Mumbai)

Database Tables

Table 1: `workouts`

```sql
create table workouts (
  id uuid default gen_random_uuid() primary key,
  day_key text unique not null,
  checked_in boolean default false,
  exercises jsonb default '{}',
  created_at timestamp default now()
);
```

- `day_key` format: `YYYY-M-D` (e.g. `2026-3-18`) — no zero padding
- `checked_in` — true when user clicks "Mark Today as Done"
- `exercises` — JSON object like `{"2026-3-18_0": true, "2026-3-18_1": true, ...}` — keys are `daykey_exerciseIndex`

Table 2: `streak_meta`

```sql
create table streak_meta (
  id int primary key default 1,
  streak int default 0,
  total_workouts int default 0,
  updated_at timestamp default now()
);

insert into streak_meta (id) values (1);
```

- Single row, id always = 1
- `streak` — current consecutive day streak
- `total_workouts` — total days checked in ever

Current Data in Supabase (as of March 2026)

- `2026-3-18` — checked_in: true, all 7 exercises done

- `2026-3-19` — checked_in: true, all 7 exercises done
- `2026-3-20` — checked_in: true, all 7 exercises done
- `2026-3-21` — checked_in: true, all 7 exercises done
- streak_meta — streak: 4, total_workouts: 4

Spotify Configuration

Client ID: 1a36710bc8cc4f228d4a9825fa9acad2
Client Secret: f8edae560c2542c497d9a3dbad0279f8
Auth Flow: Implicit Grant (response_type=token) — token returned in URL hash
Redirect URI registered: https://nidhithakur-24.github.io/daily-ritual/
Scopes: user-read-playback-state, user-modify-playback-state, user-read-currently-playing
App Status: Development mode — only whitelisted users can log in

Spotify Status

- Login button exists in app under Music tab
- On Chrome: should work (untested successfully)
- On Edge: blocked by Edge tracking prevention — shows "Content blocked" error
- Token stored in localStorage with expiry
- Polling every 5 seconds for now playing state
- Play/Pause/Skip/Previous controls built
- 4 preset workout playlists (Beast Mode, Yoga, Morning Motivation, Mood Booster)

App Structure

Single File Architecture

Everything lives in one file: `index.html`
~930 lines total
Structure: HTML → CSS (in `<style>`) → JavaScript (in `<script>`)

Tabs / Panels

1. **Today** (`panel-workout`) — daily exercises, progress ring, check-in button
2. **Tracker** (`panel-tracker`) — streak stats, calendar, total workouts
3. **Nutrition** (`panel-food`) — food guidance, hydration tips

4. **Music** (`panel-spotify`) — Spotify login, now playing, playlist launcher

Bottom Navigation

4 buttons: Today (🧘) · Tracker (🗓️) · Nutrition (🥗) · Music (🎵)

Features Built

1. Daily Workout Plans

- 7 unique plans — one per day of the week (Sunday through Saturday)
- Each plan has 7 exercises
- Exercise types: yoga, strength, breathe, cardio
- Each exercise has: name, type, duration, reps, description, tip, viz key

2. Exercise Visualizations

- Custom SVG stick figure diagrams for every exercise
- ~35 unique SVG visualizations
- Stored in `getViz(type)` function as a lookup object
- SVGs are inline, dark background (`#0d0d0d`), colored by exercise type

3. Exercise Timer

- Per-exercise countdown timer
- Circular arc progress ring (SVG)
- Color shifts: green → yellow → orange → red as time runs out
- Start / Pause / Resume / Reset controls
- Auto-marks exercise as done when timer hits 0
- Vibrates phone on completion (navigator.vibrate)
- Timer duration parsed from exercise duration string (e.g. "3 min" → 180 seconds)

4. Progress Ring

- SVG circular ring in hero section
- Shows X/7 exercises completed today
- Animates smoothly as exercises are checked off

5. Streak System

- Streak stored in Supabase streak_meta table
- Loaded from cloud on every app open
- `_streakFromCloud` flag prevents local recalculation from overwriting cloud value
- `recalcStreak()` only runs when user actively checks in a new day
- If today not checked in, counts backwards from yesterday

6. Calendar

- Monthly calendar view in Tracker tab
- Green = checked in day
- Red = missed past day
- Blue border = today
- Renders AFTER Supabase loads (critical — prevents showing red before data arrives)

7. Cloud Sync

- All data saved to Supabase on every state change
- localStorage (`workout_ritual_v3`) used as instant cache on load
- Sync status bar shows: spinning (loading) → green (synced) → red (offline)
- Offline fallback — uses localStorage if Supabase unreachable
- Toast notifications for save confirmations

8. Migration Banner

- Detects old localStorage key `workout_state_v2`
- Shows migration banner if old data found and cloud is empty
- One-click migration pushes all historical data to Supabase

9. PWA Support

- Web App Manifest (inline data URI)
- Apple touch icon
- Favicon
- Theme color: `#c8f55a` (lime green)
- `apple-mobile-web-app-capable` meta tag

- Works as installable app on Android (Chrome → Add to Home Screen) and iOS (Safari → Add to Home Screen)
- Custom icon: black square with concentric lime green circles

10. Nutrition Tab

- Foods to eat freely (5 items with descriptions)
- Foods to limit (5 items with descriptions)
- Hydration guidance
- Not a strict diet — lifestyle guidance only

11. Motivational Quotes

- 7 quotes, one per day of week
- Includes one Korean phrase (강한 몸은 강한 마음에서 시작된다)
- Displayed in serif font quote card

12. Daily Check-in Flow

- All 7 exercises must be ticked → "Workout Complete" banner appears
- "Mark Today as Done" button → saves to Supabase → streak increments
- Button disabled after check-in with "Logged! See you tomorrow 🔥 "

Design System

Colors (CSS variables)

```css
```

```
--bg: #0d0d0d
--surface: #161616
--card: #1e1e1e
--border: #2a2a2a
--accent: #c8f55a      /* lime green — primary brand */
--accent2: #ff6b35     /* orange — strength exercises */
--accent3: #7eb8f7     /* blue — yoga exercises */
--text: #f0f0f0
--muted: #888
--soft: #555
--danger: #ff4757      /* red */
--warm: #ffd166        /* yellow — streak count */
```

Typography

- Headings: DM Serif Display (serif, italic accents)
- Body: DM Sans (sans-serif)
- Dark theme throughout — no light mode

JavaScript Architecture

Key State Object

javascript

```javascript
let state = {
  streak: 0,
  totalWorkouts: 0,
  checkedDays: {},        // {"2026-3-18": true, "2026-3-19": true}
  todayExercises: {},     // {"2026-3-18_0": true, "2026-3-18_1": true}
  lastCheckin: null,
  _streakFromCloud: false // flag to prevent recalcStreak overwriting cloud value
};
```

Key Functions

- `loadState()` — async, loads from Supabase, falls back to localStorage
- `saveState()` — async, saves to both localStorage and Supabase
- `renderExercises()` — renders exercise list for today's plan
- `renderTracker()` — renders stats and calendar

- `renderCalendar()` — renders monthly calendar grid
- `updateProgress()` — updates progress ring and done count
- `checkInToday()` — marks today complete, increments streak, saves
- `recalcStreak()` — recalculates streak from checkedDays (skipped if _streakFromCloud)
- `getToday()` — returns date string in `YYYY-M-D` format (no zero padding)
- `getTodayPlan()` — returns workout plan for current day of week
- `timerToggle(i)` — start/pause timer for exercise i
- `timerFinished(i)` — handles timer completion, auto-marks exercise done
- `spInit()` — initializes Spotify, handles OAuth callback from URL hash
- `spotifyLogin()` — redirects to Spotify authorization page
- `spGetPlayback()` — fetches current playback state from Spotify API

Supabase Helper Functions

```javascript
dbGet(table, match)    // GET single row with filter
dbGetAll(table)        // GET all rows
dbUpsert(table, data)  // POST with merge-duplicates prefer header
```

Timer State

```javascript
const timers = {};
// timers[i] = { total, remaining, running, interval }
```

Known Issues / Bugs Fixed During Development

1. **`✓` showing as raw text on buttons** — Fixed by using plain text strings in JS innerHTML instead of HTML entities
2. **Streak showing 0 on refresh** — Fixed by `_streakFromCloud` flag and only running `recalcStreak()` after user check-in
3. **Calendar rendering red before data loads** — Fixed by calling `renderCalendar()` only after `loadState()` completes

4. **Exercise progress resetting to 1/7 on refresh** — Fixed by merging all exercise rows from Supabase on load
5. **Duplicate Supabase constants** — Fixed in final clean rebuild
6. `getViz()` **outside** `<script>` **tag** — Fixed in final rebuild, entire file is now properly structured
7. `Unexpected end of input` **syntax error** — Was missing closing `}` on `getViz()` function

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What Was NOT Completed

1. Spotify Integration (partially built, not fully working)

- **What's built:** Login button, OAuth flow, Now Playing widget, play/pause/skip controls, 4 playlist launchers, token storage
- **Problem:** Edge browser blocks the OAuth redirect with "Content blocked" error due to tracking prevention
- **On Chrome:** Should work but never confirmed successfully end to end
- **Root cause:** App is in Spotify Development Mode — only whitelisted emails can log in
- **To fix properly:** Either get Spotify Extended Quota approval, or switch to a backend proxy approach, or test specifically on Chrome with tracking prevention disabled

2. Multi-user Authentication

- **Planned:** Supabase Auth with email/password login
- **Why not built:** Project concluded before implementation
- **Current state:** Single user only. Anyone with the URL reads/writes to same database.
- **To implement:** Add Supabase Auth, add `user_id` column to both tables, add Row Level Security (RLS) policies, build login/signup UI

3. Row Level Security (RLS)

- Currently disabled on both Supabase tables
- Anyone with the anon key can read/write all data
- Safe for now because only one user knows the URL

4. Spotify for Multiple Users

- Currently Client ID is hardcoded

- For multi-user: each user would need their own Spotify OAuth session
- Needs auth system first

File Structure

```
daily-ritual/
└── index.html          (entire app — ~930 lines)
```

No other files. No package.json. No build process. No dependencies to install. Open index.html in any browser and it works.

How to Continue Development

To add multi-user auth:

1. Enable Supabase Auth in Supabase dashboard (Authentication → Providers → Email)
2. Set Site URL to https://nidhithakur-24.github.io/daily-ritual
3. Add `user_id uuid references auth.users` column to both tables
4. Enable RLS on both tables
5. Add policies: users can only read/write their own rows
6. Build login/signup UI in the app
7. Pass auth token in Supabase API headers instead of anon key

To fix Spotify:

1. Test on Chrome specifically (not Edge)
2. Make sure https://nidhithakur-24.github.io/daily-ritual/ (with trailing slash) is in Spotify Dashboard redirect URIs
3. Make sure your Spotify email is added as a user in Spotify Dashboard → User Management
4. The OAuth flow uses Implicit Grant — token arrives in URL hash (#access_token=...)
5. `spInit()` reads this hash on page load

To deploy changes:

1. Edit index.html
2. Go to github.com/NidhiThakur-24/daily-ritual

3. Click index.html → pencil icon → select all → paste new content → commit
4. GitHub Pages auto-deploys in ~60 seconds

To fix data manually (console commands):

```javascript
// Check what's in Supabase
const r = await dbGetAll('workouts');
const m = await dbGet('streak_meta', {id:1});
console.log(r, m);

// Fix a specific day
await dbUpsert('workouts', {day_key:'2026-3-22', checked_in:true, exercises:{'2026-3

// Fix streak
await fetch(`${SUPA_URL}/rest/v1/streak_meta?id=eq.1`, {
  method:'PATCH',
  headers:{...SH,'Prefer':'resolution=merge-duplicates,return=minimal'},
  body: JSON.stringify({streak:5, total_workouts:5})
});
```

Workout Plans Summary

Day	Theme	Exercise Types
Sunday	Slow start / reset	breathe, yoga, strength
Monday	New week energy	yoga, strength, breathe
Tuesday	Strength + stillness	yoga, strength, breathe
Wednesday	Midweek flow	yoga, strength, breathe
Thursday	Push through	yoga, cardio, strength, breathe
Friday	One more day	breathe, yoga, strength
Saturday	Rest but don't stop	yoga, strength, breathe

Each day has exactly 7 exercises. Exercise index 0–6.

Owner Info

GitHub: NidhiThakur-24
App URL: https://nidhithakur-24.github.io/daily-ritual
Current streak: 4 days (as of March 21, 2026)
Started: March 18, 2026

Summary of What Works Perfectly

- Daily workout plans with SVG visualizations
- Per-exercise countdown timers
- Exercise completion tracking
- Daily check-in and streak system
- Cloud sync via Supabase (survives refresh, browser change, device change)
- Monthly calendar with green/red/blue day states
- Nutrition guidance tab
- PWA — installable on Android and iOS home screen
- Custom app icon
- Motivational daily quotes
- Offline fallback via localStorage
- Migration banner for old localStorage data
- Toast notifications
- Sync status indicator

Summary of What Needs Work

- Spotify login (works in theory, browser compatibility issues)
- Multi-user authentication (not started)
- Row Level Security on Supabase (not enabled)
- Exercise progress sometimes needs manual Supabase correction when data gets corrupted during development